PAEDIATRIC ORTHO · NEET-SS / FET / INI-SS

Clubfoot and foot deformities

Part A: Congenital talipes equinovarus (CTEV, clubfoot)

1. Epidemiology and aetiology

- **About 1-2 per 1,000 live births** (higher in **Polynesians (about 7 per 1,000)**, lower in Chinese); **boys 2:1**; **bilateral in about 50%**.
- **Idiopathic (most)**: multifactorial; **genetic: PITX1-TBX4 pathway**, **family history (about 25% risk if a sibling and a parent are affected)**, **maternal smoking**, **early amniocentesis**; **oligohydramnios**.
- **Syndromic/non-idiopathic (stiffer, more recurrence)**: **arthrogryposis**, **myelomeningocele**, **diastrophic dysplasia** ("hitchhiker's thumb", **cauliflower ears**), **Larsen syndrome**, **Freeman-Sheldon**, **Möbius**, **amniotic band syndrome**, **tibial hemimelia**, **Pierre Robin**.
- **Associated: DDH** (examine the hips), **torticollis**.
- **Positional clubfoot** (intrauterine crowding): **flexible, fully correctable passively**: stretching only.

2. Pathoanatomy ("CAVE")

- **C**avus (**midfoot**: **forefoot pronated relative to the hindfoot**: **first ray plantarflexed**),
- **A**dductus (**forefoot adduction**: **medial subluxation of the navicular on the talar head**),
- **V**arus (**hindfoot**: **calcaneus inverted beneath the talus**),

- **E**quinus (**hindfoot/ankle**: **tight Achilles tendon**, plantarflexed talus).
- **Talus: neck medially and plantarly deviated (most deformed bone)**, small, plantarflexed; **navicular medially displaced, articulating with the medial side of the talar head (can reach the medial malleolus)**.
- **Calcaneus:** inverted and adducted, **rotated medially beneath the talus around the interosseous ligament** (**"calcaneus rotates under the talus"**).
- **Soft tissues: contracted posteromedial tissues** (**tibialis posterior, FHL, FDL, Achilles**, deltoid, spring ligament, **master knot of Henry**), **tight plantar fascia**; **calf atrophy (permanent, smaller foot and calf)**; **muscle fibrosis** (increased type I fibres and collagen).
- **Vascular: anterior tibial artery hypoplasia/absence** (about 85% in some studies): **caution with posterior tibial artery during surgery** (dorsalis pedis often absent).

3. Assessment

- **Pirani score** (6 signs, each 0, 0.5, 1; **max 6**):
  - **Midfoot contracture score (MFCS, max 3): curved lateral border**, **medial crease**, **lateral head of the talus coverage**,
  - **Hindfoot contracture score (HFCS, max 3): posterior crease**, **empty heel**, **rigid equinus**.
  - **HFCS over 2.5**: tenotomy likely needed.
- **Dimeglio classification** (**grades I-IV, 20-point scale**: soft-soft (benign) to stiff-stiff (very severe)).
- **Imaging:** not routinely needed (**ultrasound** occasionally); radiographs: **talocalcaneal parallelism** (**reduced Kite angle on AP: under 20 degrees** (normal 20-40 degrees); **lateral talocalcaneal angle under 25 degrees**).

4. Ponseti method (gold standard; about 95% initial correction)

- **Start early (ideally within the first 1-2 weeks of life)**; effective up to about 2 years (and even in older children with modifications).
- **Serial manipulation and long-leg (above-knee, knee flexed 90 degrees) plaster casts**, changed **weekly** (**about 5-7 casts**).
- **Order of correction: "CAVE"**:
  - **Cavus first**: **supinate the forefoot (elevate the first ray)** to align the forefoot with the hindfoot (**never pronate**: pronation worsens cavus),
  - **Adductus and varus together**: **abduct the supinated foot with counter-pressure on the lateral head of the talus** (**fulcrum = lateral talar head, NOT the calcaneocuboid joint**: Kite's error, which locks the calcaneus and prevents its correction),
  - **Hindfoot varus corrects spontaneously** as the calcaneus abducts/rotates under the talus,
  - **Equinus last**: **abduct to about 60-70 degrees (thigh-foot angle)** before addressing equinus.
- **Percutaneous Achilles tenotomy** (**about 80-90% need it**): when **dorsiflexion under 10-15 degrees** after abduction is achieved; under local anaesthetic; **final cast for 3 weeks** in about **60-70 degrees abduction and 15-20 degrees dorsiflexion**; tendon regenerates.
- **Errors: pronating the foot** (iatrogenic **rocker-bottom foot**), **thumb on the calcaneocuboid joint (Kite)**, **dorsiflexing before correcting varus/adduction** (**rocker-bottom/midfoot break**), **short-leg casts** (slip), **external rotation of the foot in the cast (lateral talar head as fulcrum required)**.
- **Bracing (foot abduction orthosis: Denis Browne bar / Steenbeek / Mitchell boots): affected foot about 60-70 degrees external rotation (abduction), normal foot about 30-40 degrees**, **shoulder-**

width bar, **dorsiflexion bend about 10-15 degrees**; **23 hours/day for 3 months, then night and nap time until about 4-5 years**.

- **Non-compliance with bracing = the commonest cause of relapse** (**relapse about 10-30%**: up to 80% with poor compliance vs about 6% with good compliance).
- **Relapse: dynamic supination** (**overactive tibialis anterior: walking on the lateral border**), **recurrent equinus**, adduction.
  - **Repeat casting ± repeat tenotomy**.
  - **Tibialis anterior tendon transfer (TATT) to the lateral cuneiform (third cuneiform)** for **dynamic supination** in a **child over about 2.5-3 years with flexible deformity and an ossified lateral cuneiform**.
- **Atypical (complex) clubfoot** (**short, fat, stiff foot; deep plantar and posterior creases; short hyperextended first toe; equinus severe**): **modified Ponseti (Ponseti complex clubfoot protocol)**: **avoid excessive abduction**, **hyperflexion of the knee**, abduct only to about 40 degrees, use the index finger on the lateral talar head and both thumbs on the plantar metatarsal heads to correct cavus.

5. Surgical management (residual/neglected/syndromic)

- **Posteromedial release (PMR, Turco/McKay/Carroll; Cincinnati incision)**: **historical for primary treatment** (stiffness, overcorrection, scarring, **pain and arthritis in adulthood**); now for **resistant/syndromic** feet.
- **Bony procedures (older children):**
  - **Lateral column shortening (Evans calcaneocuboid fusion/Lichtblau resection)** (**4-8 years**),
  - **Medial column lengthening (cuneiform opening wedge) + cuboid closing-wedge osteotomy** (residual adductus),
  - **Calcaneal osteotomy (Dwyer lateral closing wedge)** for residual hindfoot varus,

  - **Talectomy** (**arthrogryposis, rigid recurrent clubfoot**),
  - **Triple arthrodesis** (**over about 10-12 years**: salvage),
  - **Ilizarov/gradual correction (Joshi's external stabilisation system: JESS; Indian contribution)** for neglected/recurrent clubfoot in older children.
- **Neglected clubfoot in older children/adults**: **Ponseti casting still useful** in walking children (good results even up to about 10 years); **JESS/Ilizarov**, **triple arthrodesis**.

Part B: Other paediatric foot deformities

6. Metatarsus adductus

- **Commonest foot deformity in infants** (about 1 per 1,000); **forefoot adduction with a normal hindfoot (neutral/valgus) and normal ankle dorsiflexion** (no equinus: **distinguishes from clubfoot**); **"bean-shaped foot"**; associated with **DDH (about 5-10%)**.
- **Bleck classification** (heel bisector line): **normal: through the 2nd-3rd toe web**, **mild: through the 3rd toe**, **moderate: between the 3rd and 4th toes**, **severe: between the 4th and 5th toes**.
- **Flexibility** (passive correction beyond neutral) determines treatment: **flexible: observation/stretching (about 90% resolve spontaneously)**; **stiff/rigid: serial casting (best under 6-8 months)**; **surgery rarely** (**over 4-5 years** with persistent rigid deformity: **medial cuneiform opening wedge + cuboid closing wedge osteotomy**). **Avoid tarsometatarsal capsulotomy (Heyman-Herndon: high failure, arthritis)**.
- **Skewfoot (serpentine/Z-foot): forefoot adduction + hindfoot valgus** (**talonavicular lateral subluxation**): often **iatrogenic** (overcorrection of metatarsus adductus/clubfoot by casting in valgus); **medial cuneiform opening wedge + calcaneal lengthening** if symptomatic.

7. Congenital vertical talus (CVT; "rocker-bottom foot")

- **Irreducible dorsal dislocation of the navicular on the talar neck**, **talus vertical (plantarflexed)**, **hindfoot equinus and valgus**, **forefoot dorsiflexed and abducted**; **rigid rocker-bottom foot** with a **"Persian slipper" appearance**; **palpable talar head on the plantar-medial sole**.
- **About 50% associated with neuromuscular/genetic disorders**: **arthrogryposis**, **myelomeningocele**, **trisomy 13/18**, **prune belly**, **HOXD10 mutation (familial)**; **check the spine, hips, and neurological status**.
- **Diagnostic radiograph: forced (maximal) plantarflexion lateral view**: **the talar axis line stays plantar to the first metatarsal axis (the talonavicular dislocation does not reduce)** (**in congenital oblique talus, it reduces**). **Lateral talar-first metatarsal angle** (Meary) markedly abnormal; **dorsiflexion lateral**: fixed equinus of the calcaneus.
- **Treatment:**
  - **Dobbs method ("reverse Ponseti")**: **serial casting (plantarflexion and inversion: forefoot plantarflexed and inverted with counter-pressure on the talar head)** followed by **percutaneous pinning of the talonavicular joint** (after reduction) and **percutaneous Achilles tenotomy**; **± anterior tibialis transfer to the talar neck**; bracing (**like Ponseti boots**). **Best results**.
  - **Open reduction (single-stage dorsal approach or two-stage) for failed/late cases**: release of the **tibialis anterior, EHL, EDL, peroneus tertius**, dorsal talonavicular capsule; **Achilles lengthening**; **talonavicular pinning**.
  - **Older children: talectomy, subtalar arthrodesis (Grice), triple arthrodesis**.

- **Congenital oblique talus**: **flexible**; **talonavicular joint reduces in plantarflexion**: **casting/observation**.

8. Calcaneovalgus foot

- **Postural: foot dorsiflexed, lying against the anterior tibia, hindfoot valgus**: **flexible; intrauterine positioning**; commoner in **first-born girls**; **resolves spontaneously or with stretching (within weeks-months)**.
- **Differential diagnoses: CVT** (rigid: radiograph), **posteromedial bowing of the tibia** (**calcaneovalgus foot + posteromedial tibial bowing**: **spontaneous improvement; leg-length discrepancy of about 3-4 cm at maturity: epiphysiodesis/lengthening**), **myelomeningocele (L5)** (weak plantarflexors).

9. Flexible flatfoot (pes planovalgus) in children

- **Physiological in infants and toddlers** (**fat pad, ligamentous laxity**; the arch develops by about **6-10 years**).
- **Flexible: arch reconstitutes on tiptoe (Jack/Hubscher toe-raise test), hindfoot inverts on tiptoe**; usually **painless and benign**: **no treatment** (orthoses do not change the natural history).
- **Tight Achilles/gastrocnemius (Silfverskiöld test)** with symptomatic flexible flatfoot: **stretching**; **calcaneal lengthening osteotomy (Evans)** + gastrocnemius recession for **symptomatic, failed conservative care**.
- **Rigid flatfoot: tarsal coalition**, **CVT**, **inflammatory arthritis**, **peroneal spastic flatfoot**, **accessory navicular**, **neuromuscular** (cerebral palsy).

10. Tarsal coalition

- **Failure of segmentation** (autosomal dominant with variable penetrance) of tarsal bones: **fibrous (syndesmosis), cartilaginous**

(synchondrosis) or bony (synostosis); **about 1-2% (probably higher)**; **bilateral in about 50%**.

- **Calcaneonavicular (CN): commonest (about 53%)**; **presents age 8-12 years** (ossifies earlier); **talocalcaneal (TC; middle facet): about 37%**, **presents age 12-16 years**.
- **Presentation: rigid painful flatfoot**, **recurrent ankle sprains**, **peroneal spasm ("peroneal spastic flatfoot")**, **limited subtalar motion**.
- **Radiographs:**
  - **CN coalition: "anteater nose sign" (elongated anterior process of the calcaneus) on lateral**, best seen on the **45-degree internal oblique view**,
  - **TC coalition: "C sign" (continuous line of the talar dome and the sustentaculum)**, **talar beaking (dorsal talar head: traction spur, not arthritis)**, **short talar neck**, best seen on the **Harris (axial) view**; **CT (gold standard; extent of coalition as a percentage of the posterior facet)**.
- **Treatment:**
  - **Non-operative (first): cast/boot immobilisation 4-6 weeks, orthoses, NSAIDs.**
  - **CN coalition: resection with interposition of the extensor digitorum brevis (EDB) or fat** (good results).
  - **TC coalition: resection with fat/FHL interposition if the coalition is under about 50% of the posterior facet area and no significant hindfoot valgus (over about 16-21 degrees) or arthritis**; **otherwise subtalar or triple arthrodesis**; **calcaneal lengthening osteotomy** with resection for valgus.

11. Accessory navicular

- **Accessory ossicle at the medial navicular within the tibialis posterior tendon** (**about 10-15% of people**; often bilateral);

Geist/Coughlin types: I (os tibiale externum: sesamoid in the tendon), **II (synchondrosis: commonest symptomatic)**, **III (cornuate navicular: fused)**.

- **Medial midfoot pain, prominence**, associated with flatfoot.
- **Treatment: shoe modification, cast, orthoses**; surgery: **excision of the ossicle ± advancement of the tibialis posterior (Kidner procedure)** or simple excision.

12. Cavus foot in children

- **High arch, hindfoot varus**; **two-thirds have an underlying neurological cause**: **Charcot-Marie-Tooth (HMSN type I: commonest; PMP22 duplication on chromosome 17)**, **spinal cord lesions (tethered cord, diastematomyelia, syringomyelia, spinal tumours)**, **Friedreich ataxia**, **polio**, **cerebral palsy**, residual clubfoot. **Unilateral cavus = spinal cord pathology until proved otherwise (MRI spine)**.
- **CMT mechanism: weak tibialis anterior vs strong peroneus longus** (**plantarflexed first ray: forefoot-driven cavus**) and **weak peroneus brevis vs strong tibialis posterior** (**hindfoot varus**); **intrinsic weakness (claw toes)**.
- **Coleman block test:** a block under the lateral foot (first ray allowed to drop): **hindfoot varus corrects = flexible hindfoot (forefoot-driven)**: **forefoot surgery (dorsiflexion osteotomy of the first metatarsal) + plantar fascia release ± peroneus longus to brevis transfer** suffices; **does not correct = rigid hindfoot**: add **calcaneal (Dwyer/sliding) osteotomy**; **triple arthrodesis** for severe rigid deformity (avoid if possible: adjacent joint arthritis).
- **Tendon transfers: peroneus longus to brevis (balance)**, **tibialis posterior to the dorsum (weak dorsiflexion)**, **Jones transfer (EHL to the first metatarsal neck + IPJ fusion) for clawed hallux**, **Hibbs (EDL to the lateral cuneiform/cuboid)**.

13. Toe deformities

- **Curly toes** (3rd-5th; **FDL/FDB tightness**: flexion and medial rotation): **usually resolve; flexor tenotomy** if persistent (**over 3-4 years**).
- **Overlapping fifth toe (digitus minimus varus)**: **dorsal and medial**; symptomatic: **Butler procedure (racquet incision, dorsal capsule release)**.
- **Congenital hallux varus**: **associated with polydactyly, short first metatarsal, longitudinal epiphyseal bracket (delta phalanx)**.
- **Polydactyly (postaxial commonest: fifth toe; African descent)**: excise the duplicated (usually the most lateral/medial) toe at about **9-12 months**.
- **Syndactyly** of toes: no treatment needed (cosmetic).
- **Juvenile/adolescent hallux valgus**: **metatarsus primus varus**, often **family history**, **DMAA increased**; **delay surgery until skeletal maturity** (**recurrence is high when physes are open**); **hemiepiphysiodesis of the lateral first metatarsal physis** (guided growth) in younger children.
- **Köhler disease** (**osteochondrosis of the navicular**, **boys aged 4-6**; **sclerotic, flattened navicular**; **self-limiting**: **short leg walking cast 6-8 weeks** shortens symptoms).
- **Freiberg infraction** (**osteochondrosis of the second metatarsal head**, **adolescent girls**; **Smillie stages I-V**; rest, stiff-soled shoes; **dorsal closing-wedge osteotomy** (Gauthier), debridement).
- **Sever disease** (**calcaneal apophysitis**, **ages 8-12**, **active children**: **heel pain with activity**, **squeeze test**: **self-limiting**: heel cups, stretching, activity modification).

Exam pearl: **Clubfoot = CAVE; corrected in that order. Ponseti: supinate to correct cavus; abduct with counter-pressure on the lateral talar head (not the CC joint: Kite's error); Achilles tenotomy in about 80-90%; braces 23 h/day for 3 months then nights until 4-5 years. Non-compliance with bracing = commonest cause of relapse; dynamic supination = TATT to the lateral cuneiform.**

Exam pearl: **Talar neck = most deformed bone; navicular medially subluxed; anterior tibial artery often absent. Pirani score (MFCS + HFCS, max 6). Pronation during Ponseti = rocker-bottom foot.**

Exam pearl: **CVT: rigid rocker-bottom; plantarflexion lateral view: talonavicular dislocation persists (oblique talus reduces); 50% syndromic; Dobbs reverse Ponseti + TN pinning + tenotomy. Metatarsus adductus: normal hindfoot, no equinus, mostly resolves. Skewfoot = adductus + hindfoot valgus.**

Exam pearl: **Tarsal coalition: CN (anteater nose, 8-12 years, 45-degree oblique) vs TC (C sign, talar beak, 12-16 years, CT); resect with EDB interposition (CN) or fat (TC under 50% facet). Unilateral cavus = spinal MRI; bilateral cavus = CMT; Coleman block test decides hindfoot surgery. Köhler navicular (self-limiting), Freiberg 2nd MT head, Sever calcaneal apophysitis.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.